Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 10 - Sample-compelxity framework for gap-indexed bandit algorithms

Sample-efficient verifier feedback with gap indices

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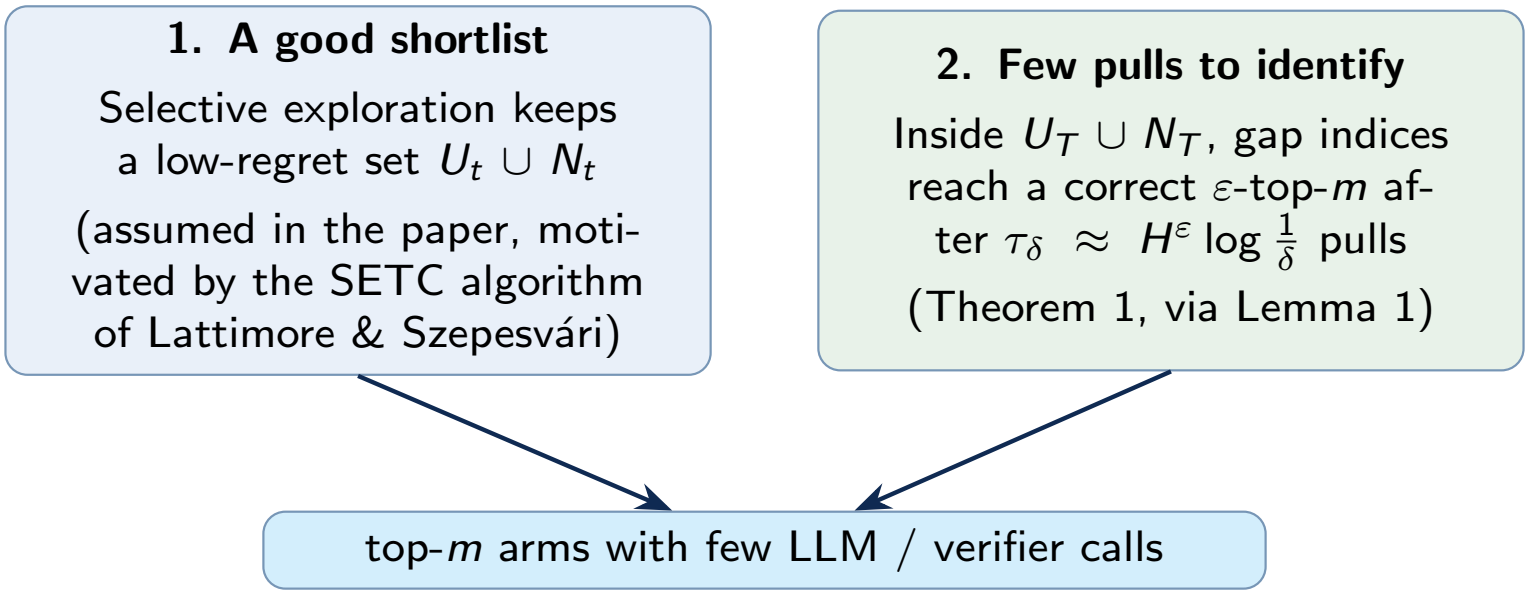
Based on LiNGIFA (AISTATS 2021) and CASE (ICML'25),

8 October 2026

Overview

1 Sample-Compelxity framework for gap-indexed algorithms

Why does gap-index algorithms work?



Today: Sample-complexity. How many pulls does correct identification need?

Setting for the analysis

- Arms: the K arms of $U_T \cup N_T$, with features x_a.
- True reward $\rho(a) = \alpha_*^\top x_a$; observations carry σ-subgaussian noise.

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- $S_m^\star$: the true top-m arms.

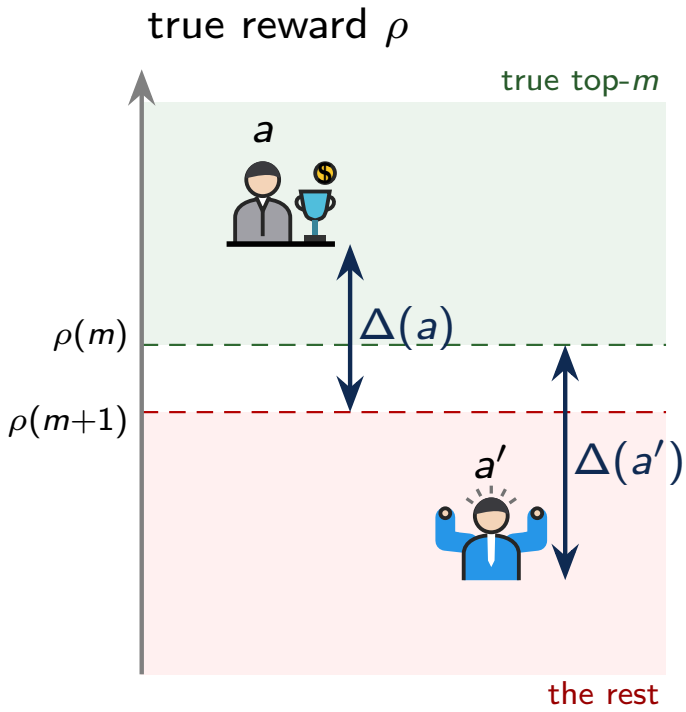
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Goal: (ε, m, δ)-PAC identification

With probability $\geq 1 - \delta$, stop and return m arms such that no other arm beats any of them by more than ε, using as few pulls as possible.

beginframeHow far is an arm from the boundary? The gap $\Delta(a)$



$$\Delta(a) = \begin{cases} \rho(a) - \rho(m{+}1) & a \in S_m^\star \\ \rho(m) - \rho(a) & a \notin S_m^\star \end{cases}$$

- Distance from a to the *other side* of the top-m boundary.
- Always $\Delta(a) \geq 0$.
- Small $\Delta(a)$: a is easy to misclassify, so it needs many pulls.

The gap index and the good event $\mathcal{E}$

Gap index and confidence width (as in the paper):

$$B_t(i,j) = \widehat{\rho}_t(i) - \widehat{\rho}_t(j) + W_t(i,j),$$

$$W_t(i,j) = C_{t,\delta}\big(\|x_i\|_{\Sigma_t} + \|x_j\|_{\Sigma_t}\big),$$

with $\Sigma_t = \sigma^2 A_t^{-1}$ and $C_{t,\delta}$ the confidence radius (β_t in Lecture 8). Note $W_t(i,j) = W_t(j,i)$.

Good event

$$\mathcal{E} = \bigcap_{t>0} \bigcap_{i,j} \Big\{ \rho(i) - \rho(j) \in \big[-B_t(j,i),\, B_t(i,j) \big] \Big\}$$

i.e. every true gap lies within W_t of its estimate.

The Lecture 8 confidence ellipsoid gives $P(\mathcal{E}) \geq 1 - \delta$. From now on we argue **on** $\mathcal{E}$.

Three facts we use again and again (on $\mathcal{E}$)

Optimism: the index never underestimates

$$\rho(i) - \rho(j) \leq B_t(i,j)$$

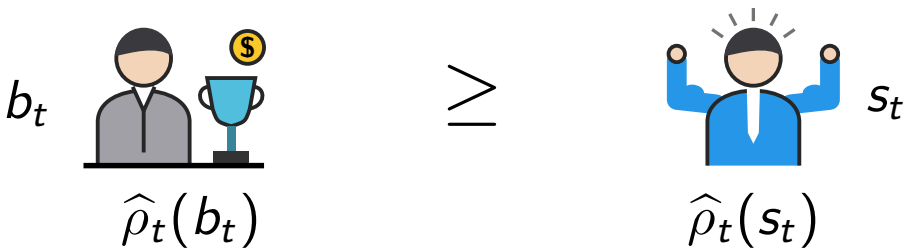
. . . but overshoots by at most $2W$

$$B_t(i,j) \leq \rho(i) - \rho(j) + 2W_t(i,j)$$

(since $\widehat{\rho}_t(i) - \widehat{\rho}_t(j) \leq \rho(i) - \rho(j) + W_t(i,j)$ on $\mathcal{E}$)

$$B_t(i,j) + B_t(j,i) = 2W_t(i,j)$$

Property 1: champions look at least as good as challengers



The swap step keeps U_t the empirical top-m of $U_t \cup N_t$, so $\widehat{\rho}_t(b_t) \geq \widehat{\rho}_t(s_t)$. Then

$$B_t(s_t, b_t) = \underbrace{\widehat{\rho}_t(s_t) - \widehat{\rho}_t(b_t)}_{\leq 0} + W_t(b_t, s_t) \;\leq\; W_t(b_t, s_t).$$

The index of the weakest link can only be positive because of **uncertainty**, not because of the estimates.

Lemma 1: the key lemma

Lemma 1 (CASE; GIFA form of Réda et al.)

On $\mathcal{E}$, for all $t > 0$,

$$B_t(s_t, b_t) \;\le\; \min\big(-(\Delta(b_t) \vee \Delta(s_t)) + 2W_t,\; 0\big) + W_t,$$

($W_t = W_t(b_t, s_t)$, $a \vee b = \max(a, b)$), i.e.

$$B_t(s_t, b_t) \le \min\big\{\, W_t,\quad 3W_t - (\Delta(b_t) \vee \Delta(s_t)) \,\big\}.$$

Reading: $\le W_t$ is Property 1; $\le 3W_t - \Delta$ says the index is *negative* unless the width is at least a third of the larger gap.

Large index $\Rightarrow$ large width: CASE never keeps working on a pair it already understands.

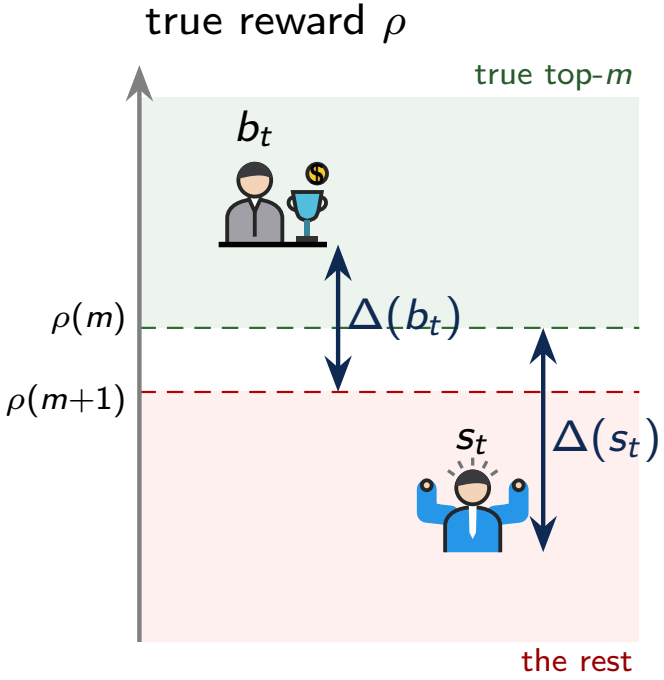
Proof of Lemma 1: four cases

It remains to show $B \le -(\Delta(b) \vee \Delta(s)) + 3W$. Shorthand: $b = b_t$, $s = s_t$, $W = W_t(b, s)$, $B = B_t(s, b)$.

	$s \in S_m^\star$ (good)	$s \notin S_m^\star$ (bad)
$b \in S_m^\star$ (good)	(iv) **both good** a bad arm hides in U_t	(i) **right order** direct
$b \notin S_m^\star$ (bad)	(ii) **wrong order** estimates inverted	(iii) **both bad** a good arm hides in N_t

Each case: compare the *true* positions of b and s with the top-m boundary.

Case (i): champion good, challenger bad



Here $b_t \in S_m^\star$ and $s_t \notin S_m^\star$, so

$$\Delta(b_t) = \rho(b_t) - \rho(m{+}1),$$

$$\Delta(s_t) = \rho(m) - \rho(s_t).$$

Notation for the true gap between two arms:

$$\Delta(i,j) := \rho(i) - \rho(j).$$

Goal (with $W_t := W_t(b_t, s_t)$):

$$B_t(s_t, b_t) \le -(\Delta(b_t) \vee \Delta(s_t)) + 3W_t.$$

Plan: (1) relate the index to the true gap $\Delta(s_t, b_t)$; (2) bound that gap using the boundary.

Case (i), step 1: from the index to the true gap

$$B_t(s_t, b_t) = -B_t(b_t, s_t) + 2W_t(b_t, s_t).$$

On $\mathcal{E}$ the index never underestimates:

$$\rho(b_t) - \rho(s_t) \leq B_t(b_t, s_t),$$

so

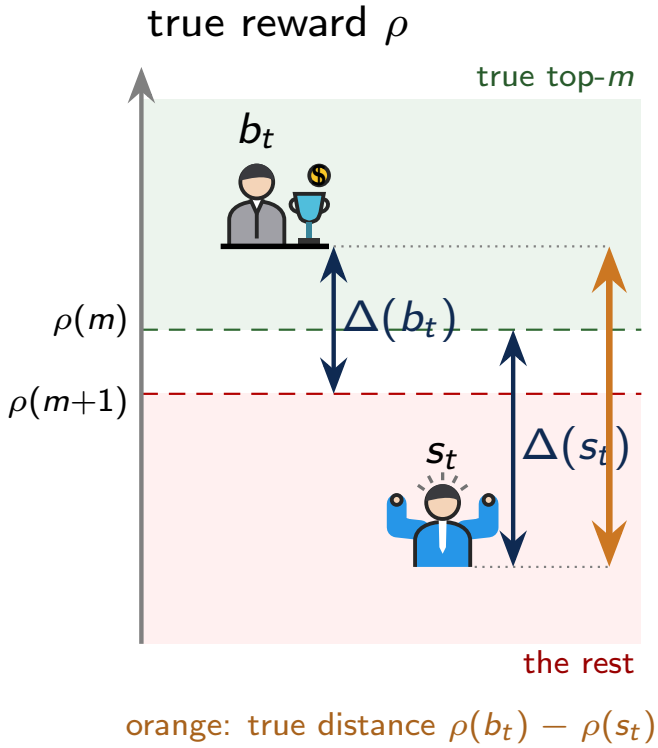
$$-B_t(b_t, s_t) \leq \rho(s_t) - \rho(b_t) = \Delta(s_t, b_t).$$

Therefore

$$\boxed{B_t(s_t, b_t) \leq \Delta(s_t, b_t) + 2W_t(b_t, s_t).}$$

On $\mathcal{E}$ the index overshoots the true gap by at most $2W$ (this is for the pair (s_t, b_t)).

Case (i), step 2: bound the true gap



orange: true distance $\rho(b_t) - \rho(s_t)$

$s_t \notin S^\star_m$, so $\rho(s_t) \le \rho(m{+}1)$:

$$\Delta(s_t, b_t) \le \rho(m{+}1) - \rho(b_t) = -\Delta(b_t).$$

$b_t \in S^\star_m$, so $\rho(b_t) \ge \rho(m)$:

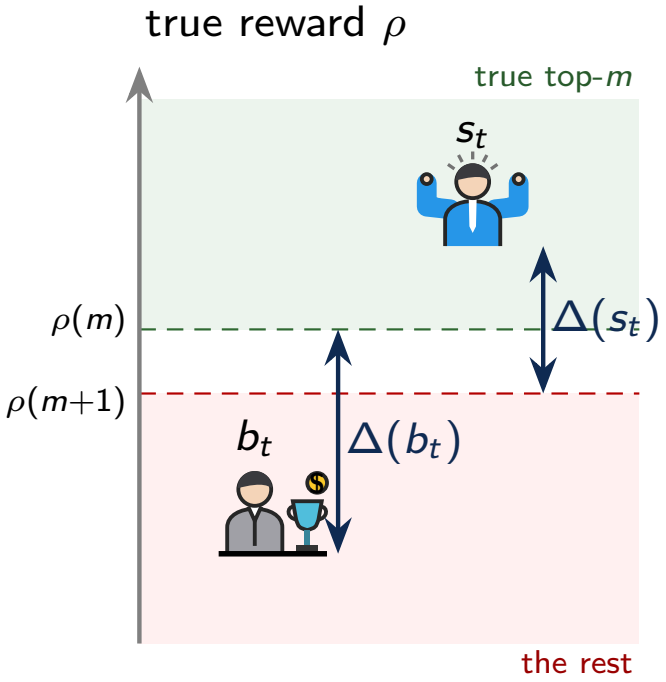
$$\Delta(s_t, b_t) \le \rho(s_t) - \rho(m) = -\Delta(s_t).$$

Hence $\Delta(s_t, b_t) \le -(\Delta(b_t) \vee \Delta(s_t))$, and with step 1:

$$\begin{aligned} B_t(s_t, b_t) &\le -(\Delta(b_t) \vee \Delta(s_t)) + 2W_t \\ &\le -(\Delta(b_t) \vee \Delta(s_t)) + 3W_t. \end{aligned}$$

b_t and s_t sit on opposite sides of the boundary, so they are at least as far apart as each one is from the boundary.

Case (ii): champion bad, challenger good



Here $b_t \notin S_m^\star$ and $s_t \in S_m^\star$, so

$$\Delta(s_t) = \rho(s_t) - \rho(m{+}1),$$

$$\Delta(b_t) = \rho(m) - \rho(b_t).$$

The estimates rank them the **wrong way round**: $\widehat{\rho}_t(b_t) \geq \widehat{\rho}_t(s_t)$, but $\rho(b_t) < \rho(s_t)$.

Estimated gap: $\widehat{\Delta}_t(i,j) := \widehat{\rho}_t(i) - \widehat{\rho}_t(j)$.

Goal (with $W_t := W_t(b_t, s_t)$):

$$B_t(s_t, b_t) \leq -(\Delta(b_t) \vee \Delta(s_t)) + 3W_t.$$

Plan: (1) flip the index with Property 1 and relate it to the true gap $\Delta(b_t, s_t)$; (2) bound that gap using the boundary.

Case (ii), step 1: flip the index, then use $\mathcal{E}$

Property 1: $\widehat{\Delta}_t(b_t, s_t) = \widehat{\rho}_t(b_t) - \widehat{\rho}_t(s_t) \geq 0$, so

$$B_t(s_t, b_t) \;\leq\; W_t \;\leq\; \widehat{\Delta}_t(b_t, s_t) + W_t \;=\; B_t(b_t, s_t).$$

Event $\mathcal{E}$: by symmetry : $B_t(i,j) + B_t(j,i) = 2W_t(i,j))$,

$$B_t(b_t, s_t) = -B_t(s_t, b_t) + 2W_t,$$

and since $\rho(s_t) - \rho(b_t) \leq B_t(s_t, b_t)$,

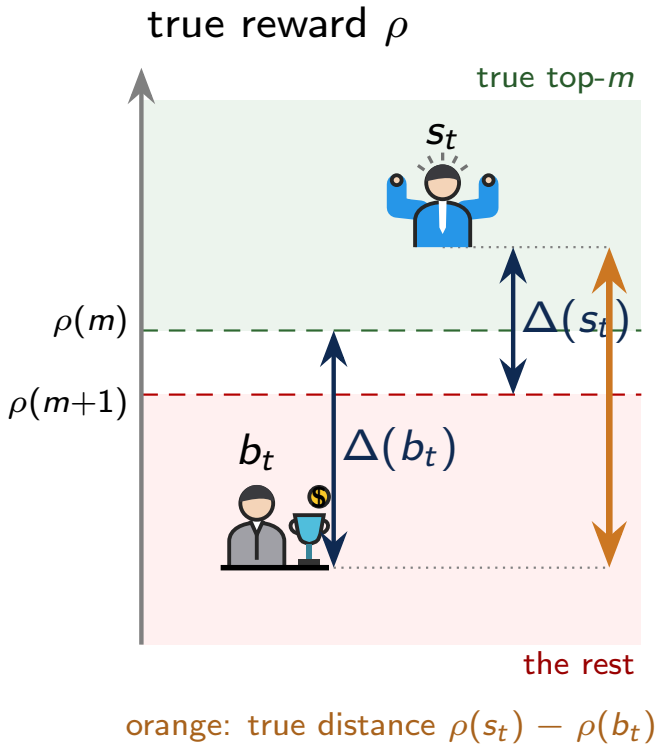
$$-B_t(s_t, b_t) \leq \rho(b_t) - \rho(s_t) = \Delta(b_t, s_t).$$

Therefore

$$\boxed{B_t(s_t, b_t) \leq B_t(b_t, s_t) \leq \Delta(b_t, s_t) + 2W_t.}$$

The estimates favour b_t, so the index "against" b_t is at most the index "for" b_t, which overshoots the true gap by at most $2W$.

Case (ii), step 2: bound the true gap



orange: true distance $\rho(s_t) - \rho(b_t)$

$b_t \notin S^\star_m$, so $\rho(b_t) \le \rho(m{+}1)$:

$$\Delta(b_t, s_t) \le \rho(m{+}1) - \rho(s_t) = -\Delta(s_t).$$

$s_t \in S^\star_m$, so $\rho(s_t) \ge \rho(m)$:

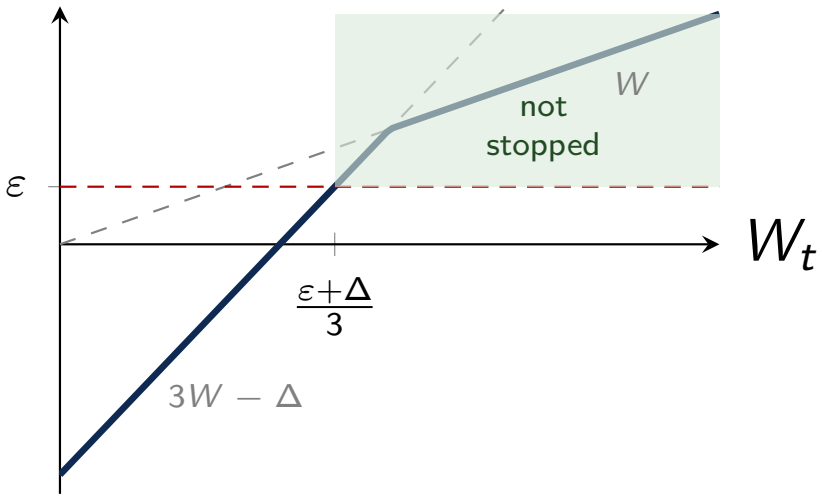
$$\Delta(b_t, s_t) \le \rho(b_t) - \rho(m) = -\Delta(b_t).$$

Hence $\Delta(b_t, s_t) \le -(\Delta(b_t) \vee \Delta(s_t))$, and with step 1:

$$\begin{aligned} B_t(s_t, b_t) &\le -(\Delta(b_t) \vee \Delta(s_t)) + 2W_t \\ &\le -(\Delta(b_t) \vee \Delta(s_t)) + 3W_t. \end{aligned}$$

Estimates say $b_t \ge s_t$, truth says s_t is ahead by at least Δ: an inverted ranking is only possible while $W_t \ge \Delta(b_t) \vee \Delta(s_t)$.

Not stopped ⇒ the pair is still uncertain



Before stopping, $B_t(s_t, b_t) > \varepsilon$.
With Lemma 1
$(\Delta = \Delta(b_t) \vee \Delta(s_t))$:

$$\varepsilon < \min\{W_t,\ 3W_t - \Delta\}$$

$$\Rightarrow \quad W_t > \max\left(\varepsilon,\ \frac{\varepsilon + \Delta}{3}\right).$$

While CASE keeps going, the pair it looks at is genuinely uncertain: its width is bounded below.

Every pull shrinks the width

Selection rule (largest variance): pull $a_t = \arg\max_{a \in \{b_t, s_t\}} \|x_a\|_{\Sigma_t}$. Then

$$W_t(b_t, s_t) = C_{t,\delta}\big(\|x_{b_t}\|_{\Sigma_t} + \|x_{s_t}\|_{\Sigma_t}\big) \le 2\, C_{t,\delta} \|x_{a_t}\|_{\Sigma_t}.$$

Pulls shrink uncertainty: if a was pulled $N_a(t)$ times, $A_t \succeq \lambda I + N_a(t)\, x_a x_a^\top$, so (Sherman–Morrison)

$$\|x_a\|^2_{A_t^{-1}} \le \frac{\|x_a\|^2}{\lambda + N_a(t)\|x_a\|^2} \le \frac{1}{N_a(t)} \quad \Rightarrow \quad \|x_a\|_{\Sigma_t} \le \frac{\sigma}{\sqrt{N_a(t)}}.$$

$$\boxed{W_t(b_t, s_t) \le \frac{2\sigma\, C_{t,\delta}}{\sqrt{N_{a_t}(t)}}}$$

Each arm has a pull budget

The upper and lower bounds on W_t meet (and $\Delta(a_t) \le \Delta(b_t) \vee \Delta(s_t)$):

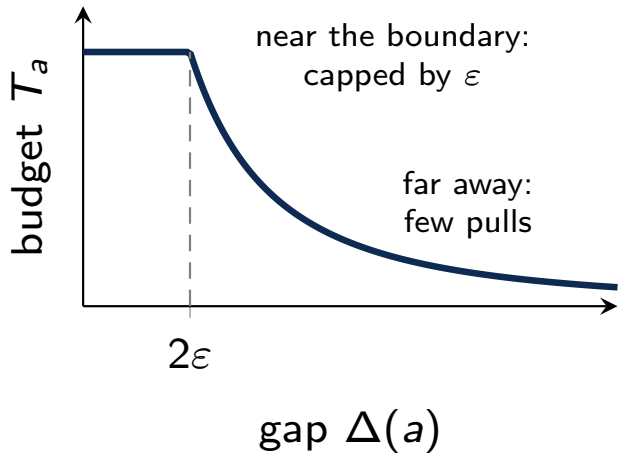
$$\frac{2\sigma C_{t,\delta}}{\sqrt{N_{a_t}(t)}} > \max\left(\varepsilon, \frac{\varepsilon + \Delta(a_t)}{3}\right) \quad \Rightarrow \quad N_{a_t}(t) < T_{a_t}(t),$$

$$T_a(t) := 4\sigma^2 C_{t,\delta}^2 \max\left(\varepsilon, \frac{\varepsilon + \Delta(a)}{3}\right)^{-2}.$$

An arm is pulled only while it is under budget, so

$$N_a(t) \le T_a(t) + 1$$

for every arm (Réda et al., Lemma 8).



Theorem 1: sample complexity of CASE

Theorem 1 (CASE; adapts Réda et al., Thm. 2)

On $\mathcal{E}$ (probability $\geq 1-\delta$), CASE returns a correct ε-top-m of $U_T \cup N_T$ (K arms) after at most τ_δ pulls, where

$$\tau_\delta \leq \inf\left\{u > 0 : \; u > 1 + H^\varepsilon \, C^2_{\delta,u} + O(K)\right\},$$

$$H^\varepsilon = 4\sigma^2 \sum_a \max\left(\varepsilon, \frac{\varepsilon + \Delta(a)}{3}\right)^{-2}.$$

Reading: H^ε sums the per-arm budgets (hard instances have many arms near the boundary); $C^2_{\delta,u} = O(\log \frac{1}{\delta} + d \log u)$ grows slowly, so $\tau_\delta \approx H^\varepsilon \log \frac{1}{\delta}$ LLM / verifier calls.

Proof of Theorem 1

Correctness (the easy part). When CASE stops, for every $b \in U_t$, $a \in N_t$,

$$\rho(a) - \rho(b) \le B_t(a,b) \le \max_{i \in U_t} \max_{j \in N_t} B_t(j,i) = B_t(s_t, b_t) \le \varepsilon.$$

Number of pulls (the main part). Before stopping, every round pulls one arm that is under budget:

$$t \;\le\; 1 + \sum_a N_a(t) \;\le\; 1 + \sum_a \big(T_a(t) + 1\big) \;=\; 1 + H^{\varepsilon}\, C_{\delta,t}^2 + K.$$

The left side grows like t, the right side only like $\log t$, so the inequality fails for large t: CASE must have stopped by the first u with $u > 1 + H^{\varepsilon} C_{\delta,u}^2 + O(K)$. □

What is proved, and what is assumed

Proved (on $\mathcal{E}$, within $U_T \cup N_T$):

- Lemma 1, Property 1 and the definitions of b_t, s_t;
- correctness at stopping, and the bound on τ_δ.

Assumed:

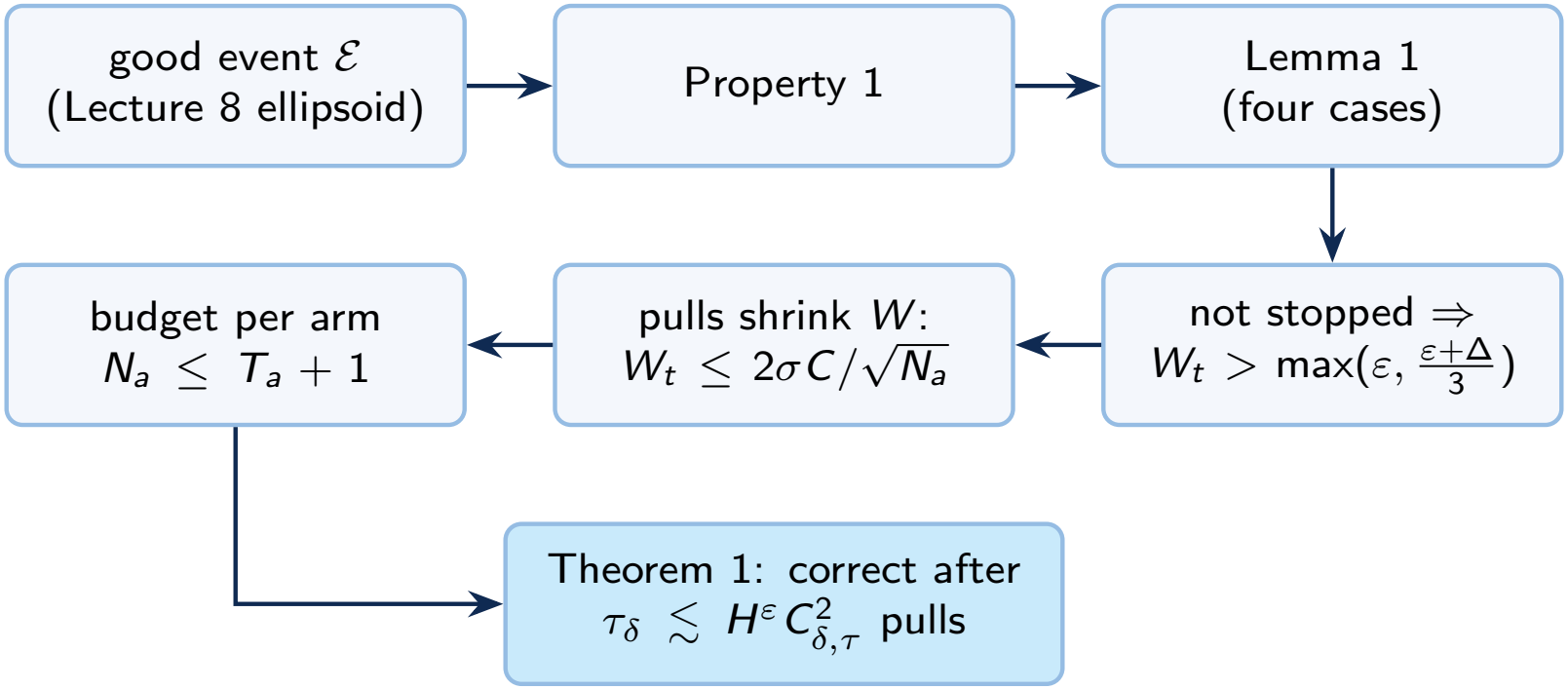
- the shortlist holds a true top-m arm when one is outside U_t (case iii);
- selective exploration makes $U_T \cup N_T$ low-regret (pillar 1, motivated by SETC; a full regret bound is left to future work);
- the swap step keeps Property 1.

The price of the shortlist: CASE vs LinGIFA

	LinGIFA (Réda et al.)	CASE
challengers	all of U_t^c	shortlist $N_t \subset U_t^c$
comparisons per round	$m(K-m)$	$m\,\lvert N_t \rvert$
case (iii)	automatic	needs $b^\star \in N_t$
guarantee	all K arms	within $U_T \cup N_T$

CASE trades a global guarantee for much cheaper rounds, which is what makes exponentially many arms (exemplar subsets) tractable.

The proof in one picture



Large index $\Rightarrow$ large width $\Rightarrow$ arm under budget $\Rightarrow$ few pulls.

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